

Simulation-Based Inference

Exercise 6

1. MCMC and Variational Inference

1. Write a Stan program for Bayesian linear regression on data (x_i, y_i) , $i = 1, \dots, N$, with the following model:

$$\begin{aligned} y_i &\sim \mathcal{N}(\alpha + \beta x_i, \sigma), \\ \alpha &\sim \mathcal{N}(0, 10), \quad \beta \sim \mathcal{N}(0, 10), \quad \sigma \sim \text{Half-Cauchy}(0, 5). \end{aligned}$$

2. Simulate $N = 100$ observations from the model above using true parameter values $\alpha = 1.0$, $\beta = 2.5$, and $\sigma = 1.0$.
3. Compile the Stan model using a programming language of your choice. Run Hamiltonian Monte Carlo with 4 chains, 1000 warmup + 1000 sampling iterations. Report posterior means and 95% credible intervals for α , β , and σ .
4. Using the same compiled model, run variational inference (`variational()`) in Stan (e.g. 20,000 iterations, 1000 output samples). Report the variational posterior means and standard deviations for α , β , and σ .
5. Plot marginal posterior densities from MCMC and VI side by side for each parameter. Compare their locations, spreads, and computational times.